Project Title	**Project Title: U.S. Airline Performance & Delay Analysis ✈️📊**
Skills take away From This Project	Welcome to your capstone project! This project challenges you to analyze a large dataset of U.S. domestic flights to uncover patterns in delays and cancellations, evaluate airline and airport performance, and provide data-driven insights. ● Advanced SQL Querying (joins, aggregations, filtering complex data) ● Business Intelligence with Power BI/Tableau (dashboard creation, data storytelling) ● Key Performance Indicator (KPI) Development & Analysis ● Data Cleaning, Transformation, and Integration ● Exploratory Data Analysis (EDA) ● Performance Benchmarking ● Analytical Report Writing & Presentation
Domain	Aviation Analytics / Transportation Management

Problem Statement:
Flight delays and cancellations are significant issues in the U.S. aviation industry, impacting passengers, airlines, and the economy. Your task is to perform an in-depth analysis of historical flight data to identify key drivers of these disruptions, assess performance, and propose actionable insights.

Business Use Cases:
The project's solution can be implemented in various scenarios, including:
● Analyzing the primary causes and patterns of flight delays and cancellations for strategic operational adjustments.
● Benchmarking the on-time performance, delay severity, and cancellation rates of different airlines for competitive analysis and improvement initiatives.
● Evaluating the operational performance of various U.S. airports to identify bottlenecks and areas for infrastructure or process improvement.
● Investigating how factors like time of day, day of week, month, and specific routes affect flight operations to optimize scheduling and resource allocation.
● Providing data-driven recommendations to stakeholders (airlines, airports, regulatory bodies) to enhance passenger experience and operational efficiency.

Approach:

Phase 1: Project Setup & Data Ingestion

- Set up your SQL database environment.
- Create appropriate table structures for flights, airlines, and airports data. Consider data types carefully.
- Import the data from the CSV files into your SQL database.
- Perform initial verification (e.g., row counts, sample data review) to ensure data is loaded correctly.

Phase 2: Data Cleaning, Preparation & Integration (SQL)

- **Time/Date Handling:** Convert text-based time and date fields (e.g., SCHEDULED_DEPARTURE) into proper datetime formats. This may involve creating new columns. Think about how to combine year, month, day, and HHMM time strings.
- **Missing Values:** Investigate and handle missing values (NULLs) in key columns (e.g., delay columns, cancellation reason). Decide on an appropriate strategy (e.g., replace with 0 where logical, leave as NULL, or flag).
- Data Enrichment: o Create a more descriptive CANCELLATION_REASON_DESC column based on the CANCELLATION_REASON code.

o Consider creating a FLIGHT_DATE column for easier time-based analysis.

- **Integration:** Create a unified analytical dataset (e.g., a comprehensive SQL View or a new table) by joining flights data with airlines and airports (for both origin and destination) to include descriptive names and locations.

Phase 3: Exploratory Data Analysis (EDA) & KPI Definition (SQL)

- Using SQL queries on your integrated dataset, explore: o Overall flight volumes, cancellations (total, by reason), and diversions.

o Basic statistics for departure and arrival delays (average, median, min, max).

o The distribution of different types of delays (airline, weather, NAS, etc.).

- **Define Key Performance Indicators (KPIs)** relevant to the project objectives. Examples include: o On-Time Performance (OTP) Rate (e.g., flights arriving within 15 minutes of schedule).

o Average Arrival/Departure Delay (in minutes).
o Cancellation Rate (%).
o Percentage contribution of each delay type.

● Perform initial aggregations to understand how these KPIs vary by: o Airline
o Origin/Destination Airport
o Month, Day of Week, Time of Day (requires robust time parsing)

Phase 4: Dashboard Development (Power BI / Tableau)

● Connect your BI tool to your SQL database (or the integrated view/table).
● Data Modeling in BI Tool: o Define necessary relationships if you load multiple tables.
o Create calculated measures/fields required for your KPIs (e.g., Total Flights, OTP Rate, Avg Delay, Cancellation Rate). *You will need to translate your KPI definitions into the BI tool's formula language (DAX for Power BI, or Tableau's functions).*

● Design and build your interactive dashboard. Consider including pages/views for: o **Overview:** Key KPIs, overall delay/cancellation reasons.
o **Airline Performance:** Comparative analysis of airlines against KPIs.
o **Airport Performance:** Comparative analysis of airports; consider a map visualization.
o **Temporal Trends:** How KPIs change by month, day of week, or hour.
● Ensure your dashboard includes filters/slicers for interactive exploration (e.g., by airline, airport, date).

Phase 5: Insight Generation & Recommendation Formulation
● Analyze your dashboards and SQL query results deeply. Ask "why?" and "so what?".
● Identify significant trends, patterns, anomalies, and correlations.
● Based on your findings, formulate 2-3 actionable and data-driven recommendations for potential stakeholders (e.g., airlines, airport authorities). Consider what changes could realistically lead to improvements.

Phase 6: Reporting & Presentation
● **Final Report:** Structure your report logically (Introduction, Methodology, Key Findings & Analysis, Dashboard Overview, Recommendations, Conclusion). Use visuals from your dashboard to support your findings.

● **Presentation:** Prepare a concise and engaging presentation summarizing your project, highlighting key insights, and showcasing your dashboard.

Results:

● Analyze the primary causes and patterns of flight delays and cancellations.
● Benchmark the on-time performance, delay severity, and cancellation rates of different airlines.
● Evaluate the operational performance of various U.S. airports.
● Investigate how factors like time of day, day of week, month, and route affect flight operations.
● Translate your findings into meaningful recommendations for stakeholders.

Data Set Explanation:

● **Content:** This dataset contains detailed records of domestic flights in the USA for 2015, including scheduled/actual times, delays, cancellations, airlines, and airports.
● Key Files:

- o flights.csv: Main flight transaction data.
- o airlines.csv: Lookup table for airline names.
- o airports.csv: Lookup table for airport details (including location).

● **Important Columns to Explore (in flights.csv):** YEAR, MONTH, DAY, DAY_OF_WEEK, AIRLINE, FLIGHT_NUMBER, ORIGIN_AIRPORT, DESTINATION_AIRPORT, SCHEDULED_DEPARTURE, DEPARTURE_TIME, DEPARTURE_DELAY, SCHEDULED_ARRIVAL, ARRIVAL_TIME, ARRIVAL_DELAY, CANCELLED, CANCELLATION_REASON, AIR_SYSTEM_DELAY, SECURITY_DELAY, AIRLINE_DELAY, LATE_AIRCRAFT_DELAY, WEATHER_DELAY, DISTANCE.

Key preprocessing steps required include:

● **Handling Time and Date:** Converting string representations of dates and times (e.g., "0855" for HHMM) into proper datetime formats to enable temporal analysis. This often involves combining year, month, day, and time components.

● **Managing Missing Values:** Addressing NULLs in critical columns such as delay durations (which might be NULL if not delayed or if cancelled) and cancellation reasons.

● **Data Type Conversion:** Ensuring all columns are of appropriate data types for analysis.

● **Data Enrichment:** Creating new descriptive columns, for example, mapping CANCELLATION_REASON codes (A, B, C, D) to their textual descriptions (Airline/Carrier, Weather, etc.).

● **Data Integration:** Joining the flights.csv data with airlines.csv and airports.csv to incorporate descriptive names for airlines and airports (origin and destination), as well as airport geographical information (city, state, latitude, longitude) into a unified analytical dataset or view.